

Experiment I: DIODE CHARACTERISTICS

Aim:

1. To study and verify the functionality of P-N Junction in Forward and Reverse Bias.
2. To study and plot the V-I characteristics of a diode.

Apparatus:

Diode, 2 Voltmeters, 470Ω Resistor, Power Supply 0-15 V, Rheostat, Connecting wires.

Key theory:

Biasing of P-N junction Diode:

Forward Bias operation.

The P-N junction supports uni-directional current flow. If the +ve terminal of the input supply is connected to P-side and the -ve terminal is connected to the N-side, then the diode is said to be in forward-biased condition. In this condition, the height of the potential barrier at the junction is lowered by an amount equal to the given forward biasing voltage. Both the holes from the P-side and electrons from N-side cross the junction simultaneously thereby decreasing the depleted region. This constitutes a forward current.

Reverse Bias operation

If the negative terminal of the input supply is connected to the P-side and the +ve terminal is connected to the N-side then the diode is said to be reverse-biased. In this condition, an amount equal to the reverse biasing voltage increases the height of the potential barrier at the junction. Both the holes on the P-side and electrons on the N-side tend to move away from the junction thereby increasing the depleted region. However the process cannot continue indefinitely, thus a small current called reverse saturation current continues to flow in the diode. This current is negligible; the diode can be approximated as an open-circuited switch it offers a very high resistance called reverse resistance.

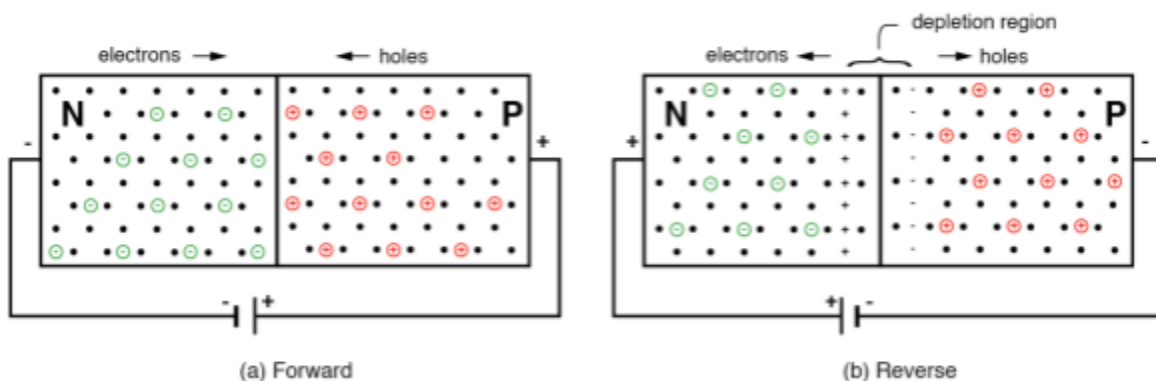


Fig 1: P-N Junction in forward and reverse biases.

Set up and procedure:

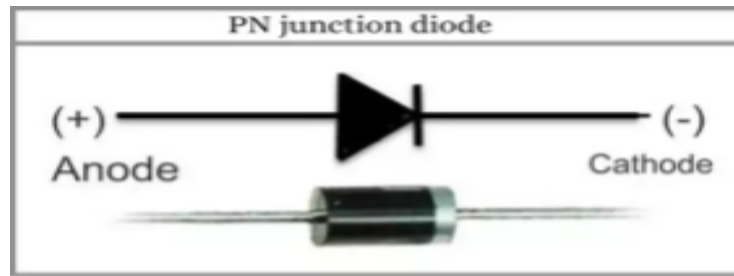


Fig 2: P-N Junction Polarity.

1. Connect the circuit shown in Figure 3. Refer to Figure 2 for the polarity of the diode.

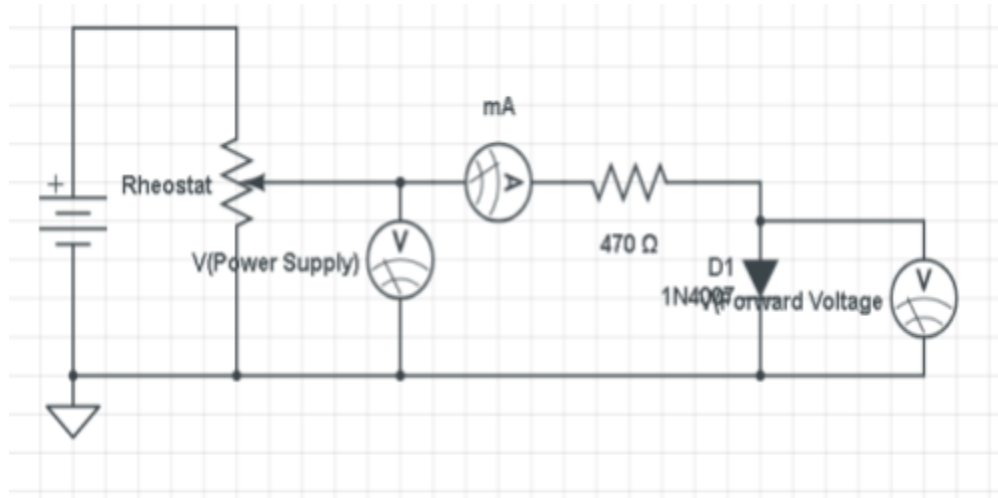


Fig 3: Forward-bias Circuit.

2. Using the rheostat, vary the power supply voltage V_s in steps of 0.1 V. Once the current starts increasing vary V_s in steps of 0.02V and note down the corresponding readings V_f and I_f .
3. Tabulate different forward currents obtained for different forward voltages.
4. Sketch your obtained results on graph paper (V_f on the X-axis, and I_f on the Y-axis). The curve obtained represents the diode characteristics in the forward region.
5. Connect the circuit shown in Figure 4.
6. Vary V_s gradually in steps of 1V from 0V to 15V and note down the corresponding readings V_r and I_r .
7. Tabulate different reverse currents obtained for different reverse voltages.
8. Plot the V-I characteristics and calculate the resistance levels.
9. Compare the theoretical and practical values.

Fig 4: Reverse-bias.

Results:

Table 1: Result for Forward Bias and Reverse Bias Circuits.

Power Supply Voltage	Forward Voltage (Vf)	Current mA (If)	Power Supply Voltage	Reverse Voltage (Vr)	Current μ A (Ir)
0.2			1		
0.4			2		
0.6			3		
0.8			4		
1			5		
2			6		
3			7		
4			8		
5			9		
6			10		
7			11		
8			12		
9			13		
10			14		
12			15		

Data and analysis:

Ideal Diode V-I Curve.

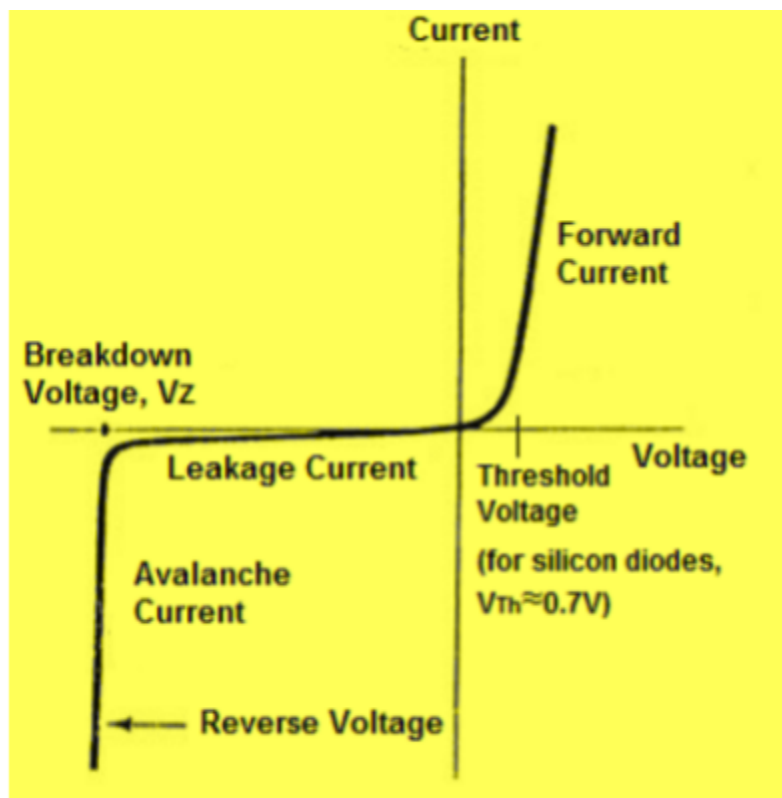


Fig 5: Ideal V-I Curve

Calculations:

- From the graphs find the following:

4a) Forward Bias of PN Junction Diode:

Cut-in Voltage $V =$

Static forward Resistance $R = V_f / I_f$

Dynamic Forward Resistance $R = \Delta V_f / \Delta I_f$

Average Resistance $R_{avg} = \Delta V_f / \Delta I_f$ (point to point)

b) Reverse Bias of PN Junction Diode:

Static forward Resistance $R = V_f / I_f$

Dynamic Forward Resistance $R = \Delta V_f / \Delta I_f$

Average Resistance $R_{avg} = \Delta V_f / \Delta I_f$ (point to point)

- From the graph determine the forward voltage at which the diode begins to conduct.

The forward voltage =

3. Determine whether the diode is a silicon or germanium diode. Show how you determined the type of diode.
4. Give one application of P-N junction diodes.